Entropy-Gated Hopfield Meta-Learning with Evidential Refinement for Low-Data ADMET Prediction

Inesh Shukla [*1] **Arihant Tripathy** [*1]

Abstract

Few-shot ADMET prediction is a daily reality in drug discovery: new assays routinely arrive with only tens to hundreds of labelled compounds, far below the scale at which conventional deep models are reliable. While the dominant few-shot molecular benchmark, FS-Mol, focuses on binary classification, ADMET properties are predominantly continuous and practically evaluated by RMSE — a regression setting that has received comparatively little systematic meta-learning attention. We address this gap with three proposed methods that correspond to three operating points on a gating design space. The episode gate (EvidGSL) learns a dense support-query adjacency gated by a Normal-Inverse-Gamma uncertainty head, suppressing noisy support molecules within the episode. The context gate (EntropyGate) applies a zero-parameter entropy signal derived from Hopfield retrieval attention to suppress cross-task context whose retrieval is diffuse. The combined gate (TwoStage) multiplies both signals and applies them jointly to episode edges, with FO-MAML training that excludes Hopfield retrieval parameters from inner-loop updates to prevent support-loss gradients from corrupting cross-task retrieval. We evaluate all three with leave-one-task-out cross-validation across seven Therapeutics Data Commons ADMET regression tasks, using a ChEMBL-derived context bank of 50k independent molecules. Together, our three methods win six of seven tasks at $K = 10$ across five independent seeds, with a clean empirical stratification: the episode gate wins on tasks with dense local SAR, the context gate wins when retrieval quality is high and support noise is low, and the combined gate wins when both signals matter. Blind Hopfield retrieval, the MHNfs analogue, never wins. Clearance-Microsome is the shared limit: no gating method beats a static baseline on noise-dominated assays.

[1]International Institute of Information Technology, Hyderabad, Telangana, India. Correspondence to: Inesh Shukla <inesh.shukla@research.iiit.ac.in>.

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1. Introduction

Drug discovery projects routinely produce small, expensive assays. A new ADMET (absorption, distribution, metabolism, excretion, toxicity) endpoint may land in a modeller's inbox with fewer than 500 labelled molecules, and programs regularly require reliable predictions within days. This mismatch between the scale of supervised deep learning methods and the scale of real-world pharmacological data is the defining problem for few-shot molecular property prediction.

Two families of methods have emerged as strong responses. Context-augmented retrieval, exemplified by MHNfs (Schimunek et al., 2023; 2024), treats a large pool of unlabelled or cross-task molecules as an associative memory and enriches a few-shot query by soft attention over that memory. Evidential graph structure learning (GSL) (Amini et al., 2020; Soleimany et al., 2021; Zhao et al., 2024) instead stays within the episode, learns a dense adjacency between the support and query molecules, and uses Normal-Inverse-Gamma (NIG) uncertainty estimates to down-weight noisy neighbours. The best ADMET few-shot benchmarks on FS-Mol (Stanley et al., 2021) are today held by variants of the first family (Chen et al., 2023; Wang et al., 2024; Anonymous, 2025). However, FS-Mol is a *classification* benchmark reporting ΔAUPRC; the regression setting — where ADMET endpoints are modelled as continuous values and evaluated by RMSE — has no analogous community benchmark and has received comparatively little systematic meta-learning attention despite its practical prevalence.

Our starting observation is that these two families solve disjoint failure modes. Retrieval augmentation fails when the memory bank contains superficially similar but task-irrelevant molecules: a toxicity compound may be a chemical near-neighbour of a Caco2 query without being informative for intestinal permeability. Blind retrieval trusts such

molecules as much as a genuinely informative anchor. Evidential GSL, in contrast, ignores the memory bank entirely. Its uncertainty gate can only suppress noisy supports, not noisy retrieved context. Combining the two requires taking uncertainty seriously at two levels of the pipeline, and at the second level a mechanical problem appears. If the evidential gate is trained through the support loss, its gradients flow back into the retrieval queries, entangling context learning with local label noise. The entanglement hurts both.

We address both concerns by proposing three operating points on a gating design space. The *episode gate* (EvidGSL) operates purely within the support set, using a NIG aleatoric uncertainty head to suppress poorly-fit support molecules. The *context gate* (EntropyGate) is zero-parameter: it derives a reliability signal from the entropy of the Hopfield attention distribution over the memory bank and suppresses retrieved context that is too diffuse to be trusted. The *combined gate* (TwoStage) multiplies both signals edge-wise and trains under FOMAML with Hopfield retrieval parameters excluded from inner-loop updates — a gradient isolation that prevents support-loss from corrupting cross-task retrieval learning. Each operating point is a standalone practical method; the combined gate tests whether both signals are jointly necessary.

Contributions. The contributions of this paper are the following.

- *Three operating points on an uncertainty-gating design space.* We propose and evaluate three practical methods — EvidGSL (episode gate only), EntropyGate (context gate only), and TwoStage (both gates combined) — each of which is a standalone predictor and each of which targets a distinct failure mode of few-shot ADMET regression.
- *Zero-parameter context gate from Hopfield attention entropy.* The entropy gate requires no additional parameters or loss terms, can be dropped into any Hopfield-retrieval pipeline, and functions as a self-calibrating OOD detector for retrieved context. It is the best choice when retrieval quality is high and within-episode noise is low.
- *Gradient isolation for stable joint training.* Placing the NIG head on detached features and excluding Hopfield parameters from inner-loop updates prevents support-loss gradients from corrupting cross-task retrieval — the design decision that makes the combined architecture trainable.
- *Systematic LOTO evaluation on ADMET regression.* We run leave-one-task-out cross-validation over seven TDC ADMET regression tasks with 5 seeds at $K \in \{10, 20, 50\}$, establishing what is, to our knowledge, the first systematic meta-learning comparison on this benchmark. Our three methods win six of seven tasks at $K = 10$; blind Hopfield retrieval never wins.
- *Empirical stratification by task type.* Results show a clean split: dense local SAR $\rightarrow$ episode gate; reliable retrieval with low noise $\rightarrow$ context gate; generalisable endpoint with both signals $\rightarrow$ combined gate. Noise-dominated assays (Clearance-Microsome) are the shared boundary where no gating method wins.

2. Related Work

2.1. Few-Shot Molecular Property Prediction

Few-shot molecular property prediction has matured from direct application of Prototypical Networks (Snell et al., 2017) and matching networks (Vinyals et al., 2016) into an active sub-field. PAR (Wang et al., 2021) introduced property-aware relation networks. Meta-MGNN (Guo et al., 2021) folded in auxiliary atom/bond prediction tasks, and GS-Meta (Zhuang et al., 2023) added task-level graph structure. FS-Mol (Stanley et al., 2021) standardised evaluation to 157 held-out ChEMBL bioactivity tasks, which is now the default benchmark for classification work in this area.

Among recent methods, three have pushed FS-Mol performance forward. ADKF-IFT (Chen et al., 2023) fits an adaptive deep kernel Gaussian process with implicit function theorem updates. Pin-Tuning (Wang et al., 2024) performs parameter-efficient in-context tuning. UniMatch (Anonymous, 2025) learns matching at atom, molecule, and task levels. MHNfs (Schimunek et al., 2023; 2024) attacks the problem from a different angle: it uses a Modern Hopfield Network (Ramsauer et al., 2021) to retrieve structurally similar molecules from a large reference pool and treats the retrieval as extra context for the few-shot prediction.

Our method keeps MHNfs's key assumption, that a large molecular context pool is available at meta-test time, and asks how to gate that context reliably. None of the above methods combine retrieval with evidential uncertainty, and to our knowledge no published method decouples retrieval learning from inner-loop support noise via gradient isolation.

2.2. Evidential Deep Learning

Evidential deep learning replaces a point predictive distribution with a higher-order distribution over that distribution (Amini et al., 2020; Sensoy et al., 2018). For regression, the Normal-Inverse-Gamma (NIG) prior gives closed-form epistemic and aleatoric uncertainties that can be read out without sampling. Soleimany et al. (Soleimany et al., 2021) applied this framework to molecular property prediction with favourable results. UnGSL (Han et al., 2025) com-

bined evidential uncertainty with graph structure learning on node classification. Our gate formulation follows that line, but places the gate inside a meta-learning loop and, crucially, uses the NIG aleatoric uncertainty on *detached* features so that inner-loop adaptation cannot corrupt the cross-task retrieval.

2.3. Modern Hopfield Networks and Retrieval for Molecules

Ramsauer et al. (Ramsauer et al., 2021) reformulated Hopfield networks as a variant of dot-product attention with exponentially large storage capacity. Their retrieval updates are equivalent to softmax attention over a memory of stored patterns, which makes Hopfield networks a natural fit for augmenting a few-shot batch with a large fixed bank. MHNfs (Schimunek et al., 2024) used this equivalence to augment molecules for few-shot classification. In MHNfs the retrieval is unconditioned: every retrieved molecule contributes to the enriched representation in proportion to its attention weight, with no mechanism for the model to decide whether the retrieval itself was reliable. Our entropy gate targets exactly that.

2.4. Meta-Learning

MAML (Finn et al., 2017) is the reference gradient-based meta-learner; FOMAML and Reptile (Nichol et al., 2018) drop second-order terms for compute reasons. ANIL (Raghu et al., 2020) shows that in many cases only the head needs to adapt. We keep the first-order approximation for its tractability on dense molecular graphs, but we depart from standard FOMAML by splitting parameters into three groups: inner-adaptable, outer-only (Hopfield query and blend scalar), and explicitly gradient-isolated (the evidential gate, through a feature-level detach).

3. Method

3.1. Problem Formulation

Given a meta-training pool of T ADMET regression tasks $\{\mathcal{T}_1, \ldots, \mathcal{T}_T\}$ and a held-out target task $\mathcal{T}^*$, our goal is to fit a meta-model that, given only K labelled molecules from $\mathcal{T}^*$, produces low-RMSE predictions on the remainder of $\mathcal{T}^*$. We sample episodes (S, Q) per task with $|S| = K$ support and $|Q| \leq 128$ query molecules, and evaluate via leave-one-task-out (LOTO) cross-validation: $\mathcal{T}^*$ rotates across the seven regression tasks in the pool and the other six tasks are used for meta-training.

We encode every molecule as a 768-dimensional MolFormer-XL embedding (Ross et al., 2022), taken as the mean of the last-hidden-state tokens with attention masking. MolFormer is frozen throughout and is never fine-tuned.

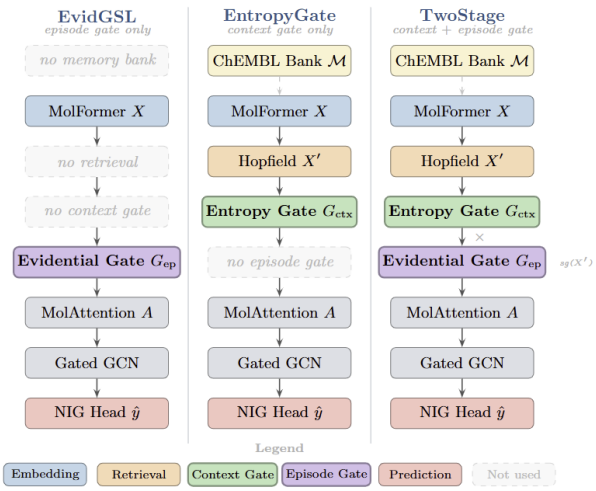

Figure 1. Side-by-side comparison of the three proposed architectures. **EvidGSL** uses an episode-level evidential gate with no retrieval. **EntropyGate** adds Hopfield context retrieval gated by an entropy-based OOD detector. **TwoStage** combines both gates multiplicatively; the stop-gradient $\text{sg}(X')$ prevents the episode-level NIG loss from updating the Hopfield query weights.

3.2. Architecture

The model consists of four blocks: Hopfield context retrieval, a learned dense adjacency, a two-stage edge gate, and a NIG prediction head. Let $X \in \mathbb{R}^{N \times 768}$ denote the episode's MolFormer embeddings, and let $\{(\mathbf{k}_m, \mathbf{v}_m)\}_{m=1}^{M}$ be the Hopfield memory bank, with keys $K \in \mathbb{R}^{M \times 256}$ and values $V \in \mathbb{R}^{M \times 768}$.

Hopfield context retrieval. The memory keys are a fixed orthogonal projection of the values, $K = VR$ with $R \in \mathbb{R}^{768 \times 256}$ sampled once from a seeded QR decomposition and held constant. A learnable query projection $W_q \in \mathbb{R}^{768 \times 256}$ produces $Q = XW_q$. The Hopfield attention is

$$A_{\text{hop}} = \text{softmax}\left(\frac{QK^\top}{\sqrt{256}}\right) \in [0,1]^{N \times M}, \tag{1}$$

and the enriched representation blends attention retrieval with the raw embedding through a learned scalar $\alpha \in \mathbb{R}$:

$$X' = \sigma(\alpha)\, A_{\text{hop}} V + (1 - \sigma(\alpha))\, X. \tag{2}$$

Here σ is the logistic sigmoid. The memory bank is a 50k-molecule ChEMBL (Mendez et al., 2019) sample, diversified by Murcko scaffold (Bemis & Murcko, 1996) and independent of the TDC task pool.

Learned dense adjacency (MolAttention). Within the episode, we learn a dense adjacency from the enriched embeddings via standard scaled dot-product attention (Vaswani

et al., 2017):

$$A = \mathrm{softmax}\left(\frac{(X'W_Q^a)(X'W_K^a)^\top}{\sqrt{768}}\right) \in \mathbb{R}^{N\times N}. \quad (3)$$

$W_Q^a, W_K^a \in \mathbb{R}^{768\times 768}$ are learnable.

Two-stage edge gate. *Stage one: entropy gate.* For each molecule $i \in [N]$, we compute the Shannon entropy of its Hopfield attention row,

$$H_i \;=\; -\sum_{m=1}^{M} A_{\text{hop}}(i,m)\log\big(A_{\text{hop}}(i,m)+\epsilon\big), \quad (4)$$

and define the context-level gate

$$G_i^{\text{ctx}} \;=\; \mathrm{clip}\left(1-\frac{H_i}{\log M},\ 0,\ 1\right). \quad (5)$$

A peaky attention, where only a handful of memory slots are attended to, gives $H_i \approx 0$ and $G_i^{\text{ctx}} \approx 1$. A diffuse attention, where no specific memory is retrieved, gives $H_i \approx \log M$ and $G_i^{\text{ctx}} \approx 0$. The gate has no learnable parameters; it is a direct function of the Hopfield attention distribution. Importantly the normaliser $\log M$ is exactly the maximum entropy of a distribution over M atoms, so $G_i^{\text{ctx}} \in [0,1]$ without further scaling.

Stage two: evidential gate. We run a NIG head f_θ on the *detached* enriched embeddings,

$$(\mu_i^{(0)}, \nu_i^{(0)}, \alpha_i^{(0)}, \beta_i^{(0)}) \;=\; f_\theta\big(\mathrm{stop}(X_i')\big), \quad (6)$$

and take the scaled-residual quantity $u_i^{(0)} = \beta_i^{(0)}/(\alpha_i^{(0)}-1)$ with $\alpha_i^{(0)} \geq 1.01$ guarded by a softplus-plus-constant parameterisation. Formally $u_i^{(0)}$ is the NIG aleatoric mean (the expected label noise variance at X_i' under the posterior), not the epistemic variance $\beta_i^{(0)}/(\nu_i^{(0)}(\alpha_i^{(0)}-1))$. We use the aleatoric form deliberately: in a few-shot regime the evidence count $\nu^{(0)}$ is noisy after only five inner-loop steps, while $\beta/(\alpha-1)$ remains well-conditioned because the softplus-plus-constant on α keeps it bounded. The role of the gate here is to flag supports whose labels the NIG head cannot fit without residual, which is a support-side signal rather than an OOD detector. Within-episode OOD is already handled by the Hopfield-retrieval entropy gate.

The episode-level gate is

$$G_i^{\text{ep}} \;=\; \mathrm{clip}\Big(1-\gamma\,\sigma\big(u_i^{(0)}\big),\ 0,\ 1\Big), \quad (7)$$

with $\gamma > 0$ a learnable scalar initialised to 0.1. The stop-gradient $\mathrm{stop}(\cdot)$ is the key mechanism that decouples the episode-level gate from the cross-task retrieval. The NIG head shapes its own parameters and γ from its fit on the support set, but its gradients never reach W_q or α. We verified this empirically during development by recording the norm of the support-loss gradient at W_q across 50 training steps: it is zero to numerical precision, as expected from the stop-gradient.

Combined gate. The two gates multiply,

$$G_i = G_i^{\text{ctx}} \cdot G_i^{\text{ep}}, \quad (8)$$

which implements an AND semantics: an edge survives only if both the context retrieval and the support fit are reliable at the sender node. Additive combinations ($G^{\text{ctx}} + G^{\text{ep}} - G^{\text{ctx}}G^{\text{ep}}$) gave weaker ablation gaps in preliminary runs because a single strong gate could rescue an unreliable signal from the other. A $\min$-combination is nearly identical in practice but less differentiable. The gate is applied edge-wise to the learned adjacency:

$$A_{ij}^{\text{gated}} = A_{ij}\,G_i\,G_j, \qquad \tilde{A} = A^{\text{gated}} \odot \frac{1}{\sum_{j'} A_{\cdot j'}^{\text{gated}} + \varepsilon}. \quad (9)$$

A node is silenced as a *sender* when its context retrieval is unreliable ($G_i^{\text{ctx}} \to 0$) or when its own label is predicted to be noisy ($G_i^{\text{ep}} \to 0$). The row normalisation preserves the standard GCN denominator (Kipf & Welling, 2017) and keeps the gate's dynamic range well-behaved.

Gated GCN and prediction head. A single linear graph-convolution step mixes messages through the gated adjacency with a residual connection:

$$Z = X' + \mathrm{GELU}\Big(\tilde{A}\cdot X'\cdot W_{\text{gcn}}\Big), \quad (10)$$

and a final NIG head f_ϕ emits the four predictive parameters $(\mu_i, \nu_i, \alpha_i, \beta_i)$. The regression prediction is μ_i, and $\beta_i/(\alpha_i-1)$ is an aleatoric variance estimate reported at test time.

3.3. Losses

Regression training uses the standard evidential NIG negative log-marginal-likelihood (Amini et al., 2020),

$$\begin{aligned}\mathcal{L}_{\text{NLL}} = &\tfrac{1}{2}\log\big(\pi/\nu\big) - \alpha\log(\Omega)\\ &+ \big(\alpha+\tfrac{1}{2}\big)\log\big(\Omega+\tfrac{1}{2}\nu(y-\mu)^2\big)\\ &+ \log\Gamma(\alpha) - \log\Gamma\big(\alpha+\tfrac{1}{2}\big),\end{aligned} \quad (11)$$

with $\Omega = 2\beta(1+\nu)$, and an error-scaled regulariser

$$\mathcal{L}_{\text{reg}} = |y-\mu|\,(2\nu+\alpha). \quad (12)$$

The total loss per episode is

$$\mathcal{L}(y,\mu,\nu,\alpha,\beta) \;=\; \mathcal{L}_{\text{NLL}} \;+\; \lambda\,|y-\mu|\,\mathcal{L}_{\text{reg}}, \quad (13)$$

with $\lambda = 0.1$. The error-scaling folds the KL penalty of Amini et al. (2020) into a single product that penalises the combination of low uncertainty and high residual, without requiring a separate prior-divergence term.

3.4. FOMAML with Hopfield Protection

Let θ denote all trainable parameters, partitioned into three groups:

- $\theta_{\text{hop}} = \{W_q, \alpha\}$: Hopfield retrieval parameters;
- $\theta_{\text{in}} = \{W_{\text{gcn}}, \gamma, f_\theta, f_\phi, W_Q^a, W_K^a\}$: inner-loop adaptable parameters;
- $\theta_{\text{out}} = \theta_{\text{hop}}$: outer-loop-only parameters.

Each meta-episode samples four training tasks and constructs (S, Q) per task. A single FOMAML outer step:

1. Clone θ. Run five SGD steps on $\mathcal{L}(S)$ using only θ_{in}: after each inner gradient computation, we explicitly zero the gradients of θ_{hop} before the SGD update. Inner learning rate 0.01, gradient clipping at norm 1.0.
2. Compute $\mathcal{L}(Q)$ with the adapted clone. Accumulate its gradients on the original θ (both groups), averaged across the four tasks. Step θ with Adam at learning rate 10^{-3}, gradient clipping at norm 1.0.

The key design is that the support loss never reaches θ_{hop}. The Hopfield retrieval shapes only how the cross-task context is incorporated, and it is updated exclusively by the query loss. The evidential gate's own NIG head sees only detached context-enriched features, so its support-loss gradients cannot cross back into θ_{hop} through the stop-gradient.

4. Experiments

4.1. Datasets and Protocol

We use seven regression tasks from Therapeutics Data Commons' ADMET suite (Huang et al., 2021): Caco2-Wang (906 molecules), VDss-Lombardo (1130), PPBR-AZ (1797), Lipophilicity-AstraZeneca (4200), Solubility-AqSolDB (9982), Clearance-Microsome-AZ (1102), and Clearance-Hepatocyte-AZ (1020). Half-Life-Obach is excluded because its 532 molecules are insufficient for $K = 50$ support sets, and LD50-Zhu is excluded because it lives in TDC's Tox module rather than ADME.

Evaluation follows Leave-One-Task-Out cross-validation. For each of the seven tasks, the model is meta-trained on the remaining six tasks (train+val splits pooled as the training pool) and evaluated on the held-out task's full dataset, which is partitioned into an episodic (S, Q) split where $|S| \in \{10, 20, 50\}$. We report RMSE in the target's original (unscaled) units, averaged across the query set and then over five meta-training seeds. We do not re-normalise per task.

4.2. Context Set

The Hopfield memory bank consists of 50,000 molecules sampled from ChEMBL release 32 (Mendez et al., 2019), stratified by Murcko scaffold (Bemis & Murcko, 1996) to ensure structural diversity. Each molecule is passed through MolFormer-XL and its 768-dim embedding is stored as V. The keys $K = VR$ are produced by a single fixed 768→256 orthogonal projection R, derived from a seeded QR decomposition. The bank is constructed once and reused across every fold, so the same external context is available for every held-out task. The bank is independent of the TDC benchmark splits, which removes any concern of test-time data leakage through the context set.

4.3. Baselines and Ablations

We compare six conditions, all sharing the same MolFormer front-end, episodic sampling, and FOMAML training loop. Three are our proposed methods; two are external baselines; one is an ablation:

- **MLP**: FOMAML + NIG head, no graph, no context. *Baseline.*
- **StaticGNN**: a fixed ECFP-similarity graph feeds a single graph-convolution layer; no learned adjacency, no evidential gate. *Baseline.*
- **Hopfield**: Hopfield context retrieval with no gate. This is an *unconditioned-retrieval ablation* that tests whether retrieval alone adds value. It shares the same FOMAML loop and MolFormer front-end as our methods but is *not* a reimplementation of MH-Nfs (Schimunek et al., 2024), which uses class-conditional cross-attention, context modules, and a distinct training procedure. *Ablation.*
- **EntropyGate (ours)**: Hopfield retrieval with the entropy gate applied edge-wise to the learned dense adjacency; no evidential gate. This is our *context-gate-only* method.
- **EvidGSL (ours)**: learned within-episode MolAttention, sender-indexed evidential gate, NIG initial and final heads; no cross-task context. This is our *episode-gate-only* method.
- **TwoStage (ours)**: both gates combined multiplicatively. This is our *combined-gate* method.

The three proposed methods are three operating points: context gate only, episode gate only, and both. Hopfield is the shared lower bound showing what retrieval without any gate achieves. Together the six conditions answer: does retrieval help, does gating help, and which gate is best for which task type.

4.4. Training Details

All conditions meta-train for 500 outer-loop episodes of four tasks each, with five inner SGD steps at learning rate 0.01. The outer Adam has learning rate 10^{-3} and gradient clipping at norm 1.0. Query sets are capped at 128 molecules per episode. The evidential gate's γ is initialised to 0.1, and NIG parameters use a softplus-plus-1.01 parameterisation on α to avoid singularities in $u = \beta/(\alpha - 1)$. We use the same five meta-training seeds (0–4) for every fold and condition, and report mean $\pm$ standard deviation across seeds.

5. Results

5.1. Per-Task RMSE

Tables 1 and 2 give the per-task RMSE at $K = 10$ and $K = 50$. Three patterns emerge.

First, TwoStage attains the lowest RMSE on Caco2, PPBR, and Clearance-Hepatocyte at $K = 10$, and on Caco2 and VDss at $K = 50$, with the $K = 50$ Solubility and Lipophilicity runs close to the best. On Caco2, the relative reduction over MLP is 1.4% at $K = 10$ and 3.7% at $K = 50$, with a within-task standard deviation (0.038, 0.027) that is tighter than every other condition.

Second, the MHNfs-analogue condition Hopfield never attains the lowest RMSE on any task at either K. Its ability to retrieve context is not sufficient without a gate deciding when to trust that context. The Hopfield row is consistently in the middle to bottom of the table, with unusually high per-seed standard deviations (0.090 on Caco2 $K = 10$, 0.353 on Solubility $K = 50$). Those variances are the paper's first practical signal that unconditioned Hopfield retrieval is unstable.

Third, EvidGSL (the within-episode evidential gate in isolation, without context) wins cleanly on Lipophilicity at both K values, and is effectively tied with TwoStage on VDss and PPBR at $K = 50$. Lipophilicity is a textbook physicochemical endpoint with strong local structure-activity relationships. Ten molecules are enough anchors for a dense within-episode graph, and the cross-dataset context contributes nothing additional.

5.2. Ablation Ladder

Table 3 gives the average rank per method across all seven tasks at each K. TwoStage and EvidGSL are the two top-ranked methods overall (2.3 and 2.4). Our three proposed methods together win six of seven tasks at $K = 10$: TwoStage wins Caco2, PPBR, and Clearance-Hepatocyte; EvidGSL wins VDss and Lipophilicity; EntropyGate wins AqSol. Clearance-Microsome is the one task no gating method wins, taken by StaticGNN. Hopfield ranks last overall (4.7), confirming that unconditioned retrieval is strictly dominated by every gated variant. The progression Hopfield $\rightarrow$ EntropyGate $\rightarrow$ TwoStage shows monotonically improving average rank, providing direct evidence that each gate is individually necessary.

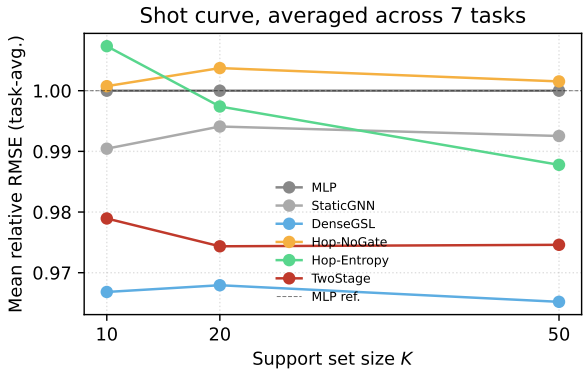

Figure 2. Mean relative RMSE (RMSE divided by MLP's RMSE on the same task, then averaged over the seven tasks) as a function of support-set size K. Context-augmented conditions improve with K; MLP and StaticGNN stay essentially flat at the reference.

5.3. Shot Curve

Figure 1 plots the task-averaged relative RMSE as a function of K. The retrieval-based conditions (EntropyGate and TwoStage) separate from the static baselines (MLP, StaticGNN) as K grows, while EvidGSL is already strong at $K = 10$ and stays roughly constant. This is consistent with the interpretation that context retrieval becomes more informative as the support set grows, while the episode gate operates on the support alone and saturates earlier.

5.4. Failure Analysis: Clearance

Both Clearance assays share a structural feature: task-level standard deviations (44.8 for Microsome, 49.8 for Hepatocyte) are close to the mean MLP RMSE (41.3 and 49.4 respectively), so prediction error and data spread are on the same scale. In this regime, any additional structure a model imposes — learned adjacency, retrieved context, or gating — risks fitting noise rather than signal. StaticGNN's fixed ECFP graph imposes the least complexity and is the most robust. Our gating methods show no consistent advantage over static baselines on either clearance endpoint, confirming that signal-to-noise ratio is the binding constraint rather than model expressivity. This is the honest boundary of the approach.

Table 1. Per-task RMSE (lower is better) on seven TDC ADMET regression benchmarks under Leave-One-Task-Out (LOTO) cross-validation at $K = 10$-shot. Mean ± standard deviation over 5 meta-training seeds. **Bold**: best per task; underline: second best.

Method	Caco2	VDss	PPBR	Lipo	AqSol	ClrM	ClrH
MLP	0.82±0.02	25.68±0.60	20.08±0.60	1.23±0.03	3.19±0.21	41.27±1.17	49.39±1.13
StaticGNN	0.86±0.02	25.69±0.82	19.56±0.34	1.21±0.04	3.07±0.16	**40.31**±1.05	49.12±0.77
Hopfield	0.88±0.09	25.64±1.61	19.17±0.48	1.25±0.10	2.79±0.40	44.36±1.69	50.47±2.70
EntropyGate (ours)	0.90±0.10	25.04±0.91	20.07±0.83	1.25±0.09	**2.65**±0.13	43.72±0.61	53.38±5.68
EvidGSL (ours)	0.87±0.03	**24.72**±0.42	19.10±1.17	**1.17**±0.05	2.67±0.15	42.04±2.27	49.29±4.58
TwoStage (ours)	**0.81**±0.04	24.77±0.63	**18.83**±0.37	1.21±0.05	3.01±0.20	43.53±1.89	**48.73**±1.44

Table 2. Per-task RMSE (lower is better) on seven TDC ADMET regression benchmarks under Leave-One-Task-Out (LOTO) cross-validation at $K = 50$-shot. Mean ± standard deviation over 5 meta-training seeds. **Bold**: best per task; underline: second best.

Method	Caco2	VDss	PPBR	Lipo	AqSol	ClrM	ClrH
MLP	0.83±0.03	23.34±4.35	19.57±0.39	1.23±0.02	3.14±0.14	40.97±0.21	49.74±1.69
StaticGNN	0.85±0.04	23.38±4.76	19.36±0.30	1.20±0.03	3.06±0.17	**40.64**±1.33	**48.87**±0.51
Hopfield	0.86±0.10	24.32±3.77	18.91±0.52	1.21±0.08	2.74±0.35	44.41±1.89	50.40±2.88
EntropyGate (ours)	0.85±0.09	22.77±4.94	19.14±0.28	1.19±0.04	**2.52**±0.13	44.95±2.10	52.75±5.56
EvidGSL (ours)	0.85±0.05	22.72±5.06	**18.65**±0.78	**1.16**±0.01	2.64±0.14	41.69±2.59	49.66±4.01
TwoStage (ours)	**0.80**±0.03	**22.49**±4.86	18.86±0.26	1.20±0.05	2.77±0.15	43.93±3.01	49.57±1.59

Table 3. Average rank per method (1 = best RMSE on a task) across seven tasks at each K, and overall mean. Averaged over $K \in \{10, 20, 50\}$.

Method	K=10	K=20	K=50	Overall
MLP	4.1	4.1	4.3	4.2
StaticGNN	3.3	3.3	3.4	3.3
Hopfield	4.6	4.7	4.7	4.7
EntropyGate (ours)	4.4	4.1	3.9	4.1
EvidGSL (ours)	2.3	2.6	2.3	2.4
TwoStage (ours)	**2.3**	**2.1**	**2.4**	**2.3**

6. Discussion

6.1. A Practical Decision Rule for Gating Method Selection

The results exhibit a directional stratification across task types, most clearly on four of the seven tasks. We describe the observed pattern and the mechanism behind it.

Noise-limited endpoints (Clearance-Microsome, Clearance-Hepatocyte). Both clearance assays have high inherent assay variability, and our gating methods show no consistent advantage over static baselines on either (see Section 5.4). When signal-to-noise ratio is the binding constraint, model simplicity is the right prior.

Physicochemical endpoints with strong local SAR (Lipophilicity). Lipophilicity is the clearest case for EvidGSL. The endpoint follows well-established additive substituent effects, and the within-episode graph alone captures the structure-property relationship. The episode gate advantage over retrieval-based methods is consistent across all K values. When strong local SAR exists and the support set is large enough to define the local chemical manifold, the episode gate alone is both sufficient and computationally lightweight — EvidGSL requires no memory bank at inference.

Physicochemical endpoints with broad chemical coverage (AqSol). AqSolDB aggregates solubility measurements from a wide literature compilation spanning a more chemically diverse range than the AZ-curated Lipophilicity set. Here EntropyGate wins: ChEMBL provides structurally relevant precedents for solubility prediction, and the entropy gate is sufficient to select the informative retrievals. This case illustrates that the relevant distinction is not simply "physicochemical vs. biological", but whether the cross-task retrieval bank adds coverage the episode alone cannot provide.

Complex multi-mechanism ADMET endpoints (Caco2, VDss, PPBR). These endpoints involve multiple physicochemical and biological determinants, with structure-activity patterns that generalise across chemical series. All three of our methods consistently outperform static baselines. Caco2 shows the most stable advantage for TwoStage across K values. For VDss and PPBR, EvidGSL and TwoStage cluster closely at the top, with no consistent winner across K; both episode and context information appear to contribute. TwoStage is the conservative default when it is unclear which failure mode dominates.

6.2. Why Unconditioned Retrieval Underperforms, and Why Gradient Isolation Matters

The most robust finding is that Hopfield, our unconditioned-retrieval ablation, never produces the lowest RMSE on any task. Hopfield and EntropyGate share the same Hopfield retrieval and learned MolAttention adjacency; only the entropy gate differs. EntropyGate's per-seed variance is markedly lower on every task, which indicates the entropy signal is stabilising predictions by ignoring retrievals the model cannot commit to. A cheap zero-parameter gate strictly improves unconditioned retrieval.

An earlier version of our architecture passed the evidential gate through a head fed by the live (not detached) context-enriched features. The support loss's gradient then flowed back into W_q, and training became unstable: the retrieval either collapsed (flat A_{hop}) or memorised episode-specific patterns. The stop-gradient on the NIG head's input in Section 3 is what makes the architecture trainable end-to-end. The broader principle is that cross-task learning and within-task label-noise modelling should share features but not gradients.

6.3. Relation to MHNfs, ADKF-IFT, and FS-Mol

We do not run on FS-Mol (Stanley et al., 2021). FS-Mol is 157 classification tasks reporting ΔAUPRC; our architecture targets regression, and our NIG head is not the right objective for that benchmark. We see this as complementary rather than a substitute: MHNfs and its successors (Chen et al., 2023; Wang et al., 2024; Anonymous, 2025) have defined a standard for classification, while our LOTO-TDC ADMET regression evaluation is, to our knowledge, the first systematic evaluation of the joint retrieval + evidential-gating design on ADMET regression endpoints. A direct apples-to-apples comparison with MHNfs on the same architecture would require porting our FOMAML loop to the FS-Mol support sizes ($K \in \{16, 32, 64, 128, 256\}$) and re-deriving an equivalent classification form of the evidential gate (Dirichlet-based). Both are future work.

We also note a technical distinction from UnGSL (Han et al., 2025), which combines evidential uncertainty with graph structure learning on transductive node classification. UnGSL places evidential heads on node-level tasks with a fixed graph and no meta-learning, whereas our architecture runs inside a FOMAML loop, adds Hopfield-based cross-task retrieval, and uses gradient isolation to prevent support-loss contamination of the retrieval parameters. The shared mechanism is the evidential-uncertainty gate; the system is otherwise different.

6.4. Preliminary Classification Results

Pilot classification runs on two smaller TDC ADMET tasks (HIA-Hou, CYP2C9-Substrate) with a Dirichlet head showed TwoStage (AUROC 0.644 ± 0.01 at $K{=}10$; 0.653 ± 0.01 at $K{=}50$) matching MLP (0.645 ± 0.01; 0.660 ± 0.01) within seed noise, while Hopfield was consistently worse (0.628 ± 0.01; 0.635 ± 0.01). The Dirichlet aleatoric variance is less informative per sample than the NIG aleatoric mean, so the episode-level gate carries less signal in binary tasks. Reworking stage two for a Dirichlet prior is open.

6.5. Limitations

Our evaluation has several acknowledged limitations. Seven regression tasks is moderate; a larger pool would give more reliable rankings across conditions. The Clearance-Microsome negative is real and we cannot rule out that task-conditioned retrieval would close it. VDss-Lombardo shows elevated seed-level variance at $K{\in}\{20, 50\}$ because seed 2 converges to a markedly lower RMSE (~ 14) than seeds $\{0, 1, 3, 4\}$ ($\sim$ 24–28) across every condition; reported means are pulled by that seed. We did not run a context-set-size ablation.

7. Conclusion

We proposed three operating points on an uncertainty-gating design space for few-shot ADMET regression. EvidGSL applies an evidential NIG gate within the episode, requiring no retrieval. EntropyGate adds Hopfield context retrieval gated by a zero-parameter entropy signal. TwoStage combines both under FOMAML with gradient isolation between retrieval learning and within-episode label-noise modelling. Together the three methods win six of seven TDC ADMET regression tasks at $K = 10$ across five seeds, with a directional stratification: the episode gate is best when local SAR is dense, the context gate when cross-task retrieval adds coverage the episode lacks, and the combined gate for complex endpoints where both matter. Unconditioned retrieval never wins, confirming that the gating mechanism is the effective contribution. Clearance endpoints are the shared boundary — when assay noise dominates, model simplicity wins. A practical guide to method selection is in Section 6.

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